

Filière : Master IAA

Projet1: Data Warehouse & BI

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Année : 2025–2026

Présentation du projet et objectifs du Data Warehouse

Contexte de projet

Ce projet vise à concevoir et implémenter un **Data Warehouse**, permettant de consolider et analyser les données hétérogènes provenant de deux sources principales : les données de ventes historiques ([Superstore Dataset](#)) et les informations de gestion de stock (base [GestionStock](#))

Objectifs principaux

- **Intégrer** des sources de données hétérogènes avec des formats et structures différents
- **Concevoir** un modèle de données optimisé pour l'analyse décisionnelle
- **Implémenter** un processus ETL robuste pour l'alimentation du Data Warehouse
- **Fournir** une base solide pour l'analyse BI et la prise de décision

Description des sources de données

- **Superstore Dataset (Format CSV)**

Fichier contenant les transactions de ventes avec 21 colonnes incluant :

- Informations commandes: Order ID, Order Date, Ship Date, Ship Mode
- Informations clients: Customer ID, Customer Name, Segment, Localisation
- Informations produits: Product ID, Category, Sub-Category, Product Name
- Mesures financières : Sales, Quantity, Discount, Profit

Volume : 9,994 enregistrements

- **Base GestionStock (MySQL)**

1. **Fournisseurs** (10 enregistrements) : Coordonnées des fournisseurs
2. **Entrepôts** (11 enregistrements) : Lieux de stockage par région
3. **Stock** (200+ enregistrements) : Quantités disponibles par produit/fournisseur/entrepôt

Relation avec Superstore : Via [Product ID](#) → [product_id](#)

La création de base de données **gestionstock** et les tableaux **entrepots**, **fournisseurs** et **stock** :

```
mysql> USE gestionstock;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_gestionstock |
+-----+
| entrepots                |
| fournisseurs             |
| stock                    |
+-----+
3 rows in set (0.07 sec)
```

```
mysql> select * from stock limit 10;
```

product_id	nom_produit	fournisseur_id	entrepot_id	quantite_stock	date_dernier_reapprovisionnement
FUR-BO-10001798	Bush Somerset Collection Bookcase	3	5	45	2024-01-15
FUR-BO-10002545	Atlantic Metals Mobile 3-Shelf Bookcases, Custom Colors	3	5	15	2024-02-17
FUR-BO-10002613	Atlantic Metals Mobile 4-Shelf Bookcases, Custom Colors	3	5	11	2024-06-03
FUR-BO-10004695	O'Sullivan 2-Door Barrister Bookcase in Odessa Pine	3	5	7	2024-04-23
FUR-BO-10004834	Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	3	5	8	2024-03-22
FUR-CH-10000454	Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	7	2	26	2024-05-14
FUR-CH-10000513	High-Back Leather Manager's Chair	7	2	14	2024-02-12
FUR-CH-10000863	Novimex Swivel Fabric Task Chair	7	2	31	2024-04-12
FUR-CH-10002331	Global Enterprise Series Seating High-Back Swivel/Tilt Chairs	7	2	17	2024-06-22
FUR-CH-10002774	Global Deluxe Stacking Chair, Gray	7	2	38	2024-03-14

```
10 rows in set (0.02 sec)
```

```
mysql> select * from entrepots
-> ;
```

entrepot_id	nom_entrepot	region
1	Entrepot Ouest	Ouest
2	Entrepot Est	Est
3	Entrepot Sud	Sud
4	Entrepot Nord	Nord
5	Entrepot Centre	Centre
6	Entrepot Ouest	Ouest
7	Entrepot Est	Est
8	Entrepot Sud	Sud
9	Entrepot Nord	Nord
10	Entrepot Centre	Centre
100	Entrepot Global	Global

```
11 rows in set (0.04 sec)
```

```
mysql> select * from fournisseurs;
```

fournisseur_id	nom_fournisseur	pays	email_contact
1	Fournitures USA	Etats-Unis	contact@fournituresusa.com
2	Mobilier Europe	France	contact@mobilierfrance.fr
3	Papeterie Monde	Allemagne	contact@papeteriemonde.de
4	Techno Supplies	Royaume-Uni	contact@technosupplies.co.uk
5	Bureautique Inc.	Canada	contact@bureautiqueinc.ca
6	Fournitures Asie	Chine	contact@fournituresasie.cn
7	Mobilier Afrique	Afrique du Sud	contact@mobilierafrique.co.za
8	Papeterie Amérique Latine	Brésil	contact@papeteriebrasil.com
9	Techno Supplies	Japon	contact@technosupplies.jp
10	Bureautique Europe	Espagne	contact@bureautiqueeurope.es

```
10 rows in set (0.02 sec)
```

Et voici le fichier **superstore** : je l'ajouter dans talend

- ▼  File delimited
 - ▼  Superstore 0.1
 - ▼  Superstore
 - ▼  Columns(21)
 -  Category
 -  City
 -  Country
 -  Customer_ID
 -  Customer_Name
 -  Discount
 -  Order_Date
 -  Order_ID
 -  Postal_Code
 -  Product_ID
 -  Product_Name
 -  Profit
 -  Quantity
 -  Region
 -  Row_ID
 -  Sales
 -  Segment
 -  Ship_Date
 -  Ship_Mode
 -  State
 -  Sub_Category

Conception du Data Warehouse

Dimensions créées :

dimclient : Information sur les clients

dimproduit : Hiérarchie produit (catégorie → sous-catégorie → produit)

dimtemps : Découpage temporel

dimfournisseur : Information sur les fournisseurs

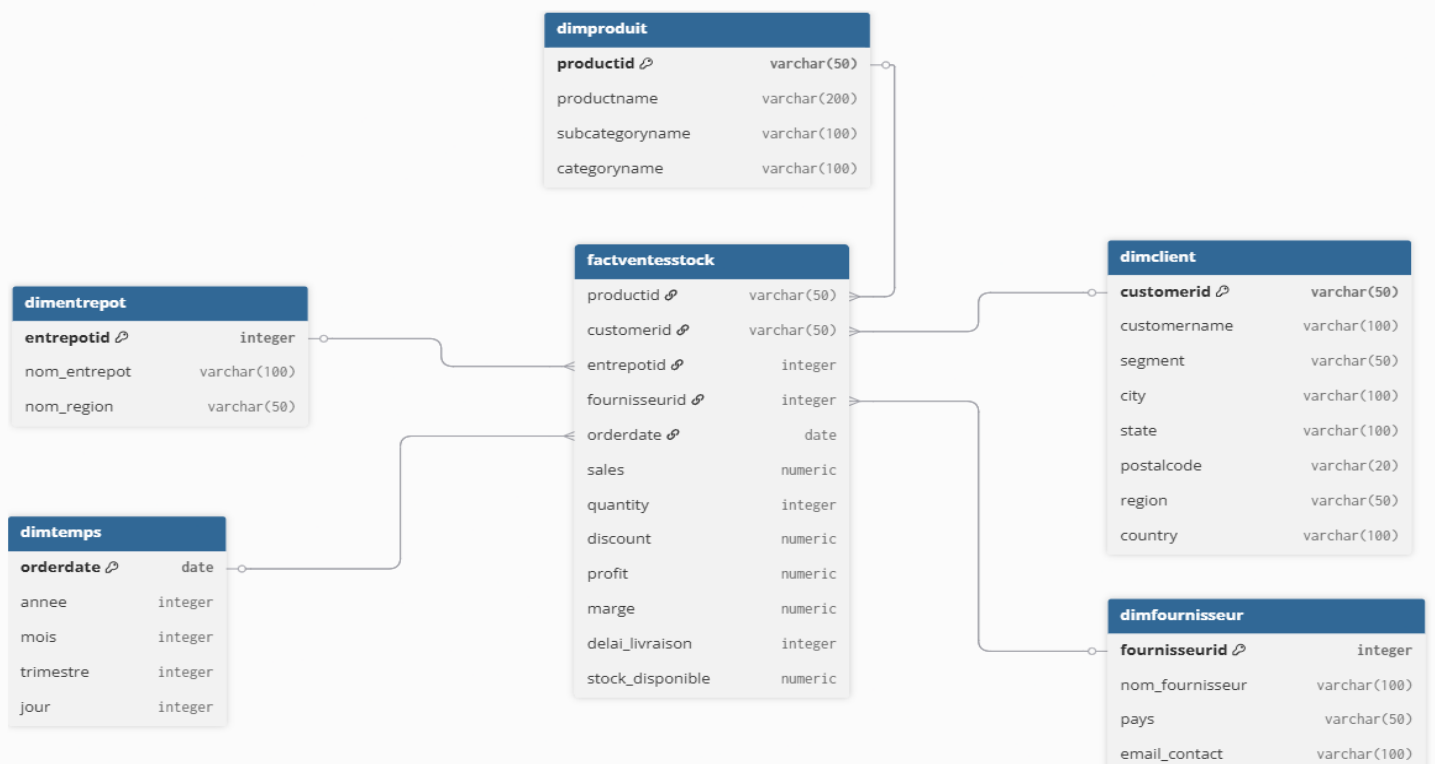
dimentrepot : Information sur les entrepôts

Table de faits principale

factventesstock :

- Agrège les ventes, stocks, et relations fournisseurs
- Granularité : Transaction produit/commande/jour
- Mesures : **sales, quantity, discount, profit, marge, stock_disponible, delai_livraison**

Voici le schéma en étoile que j'utilise :



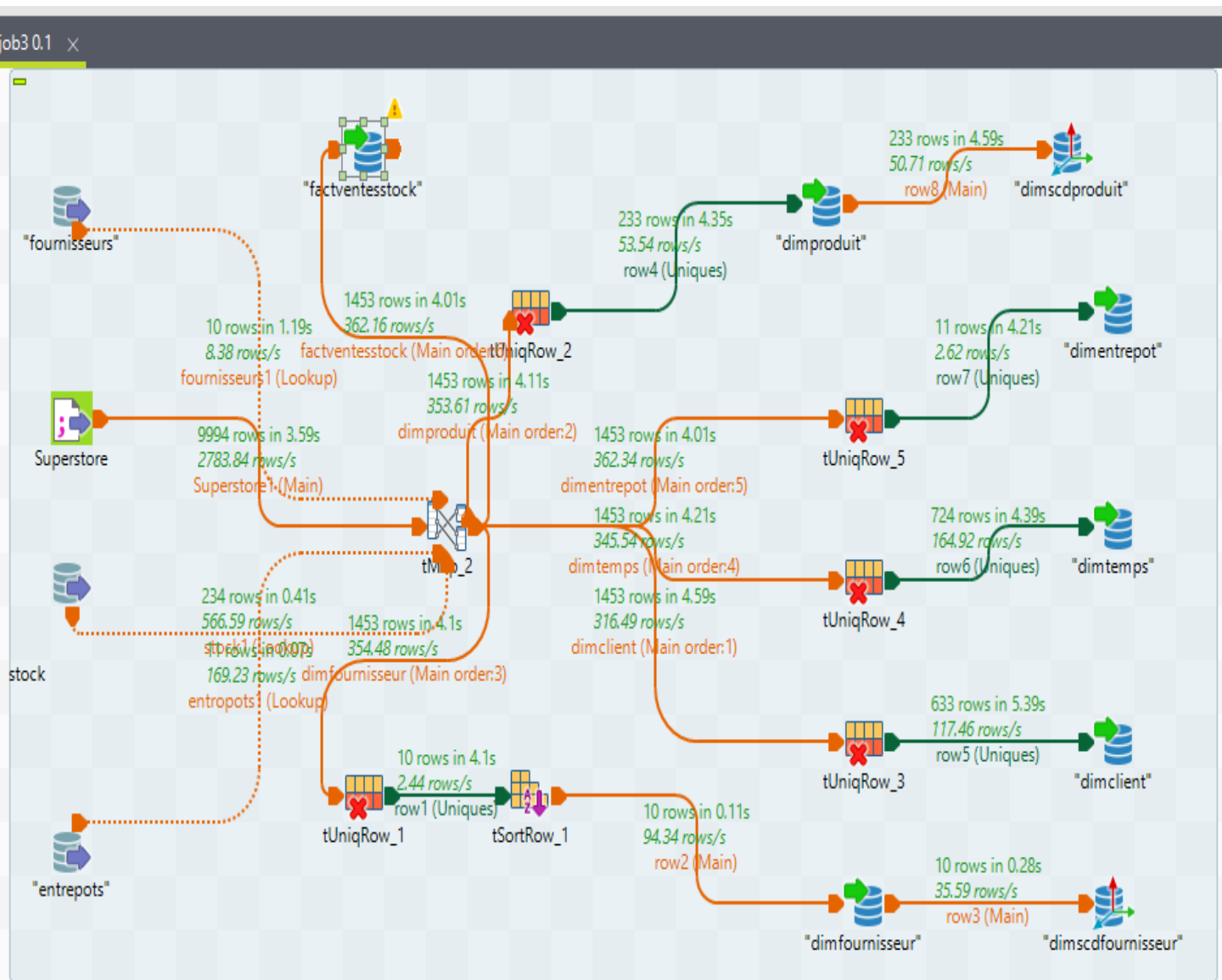
SCD (Slowly Changing Dimensions)

J'utilise Slowly Changing Dimensions de **Type 2** sur ces dimensions :

- **dimfournisseur** : Changements de coordonnées
- **dimproduit** : Changements de prix ou de catégorie
- **dimclient** : Changements de segment ou de région

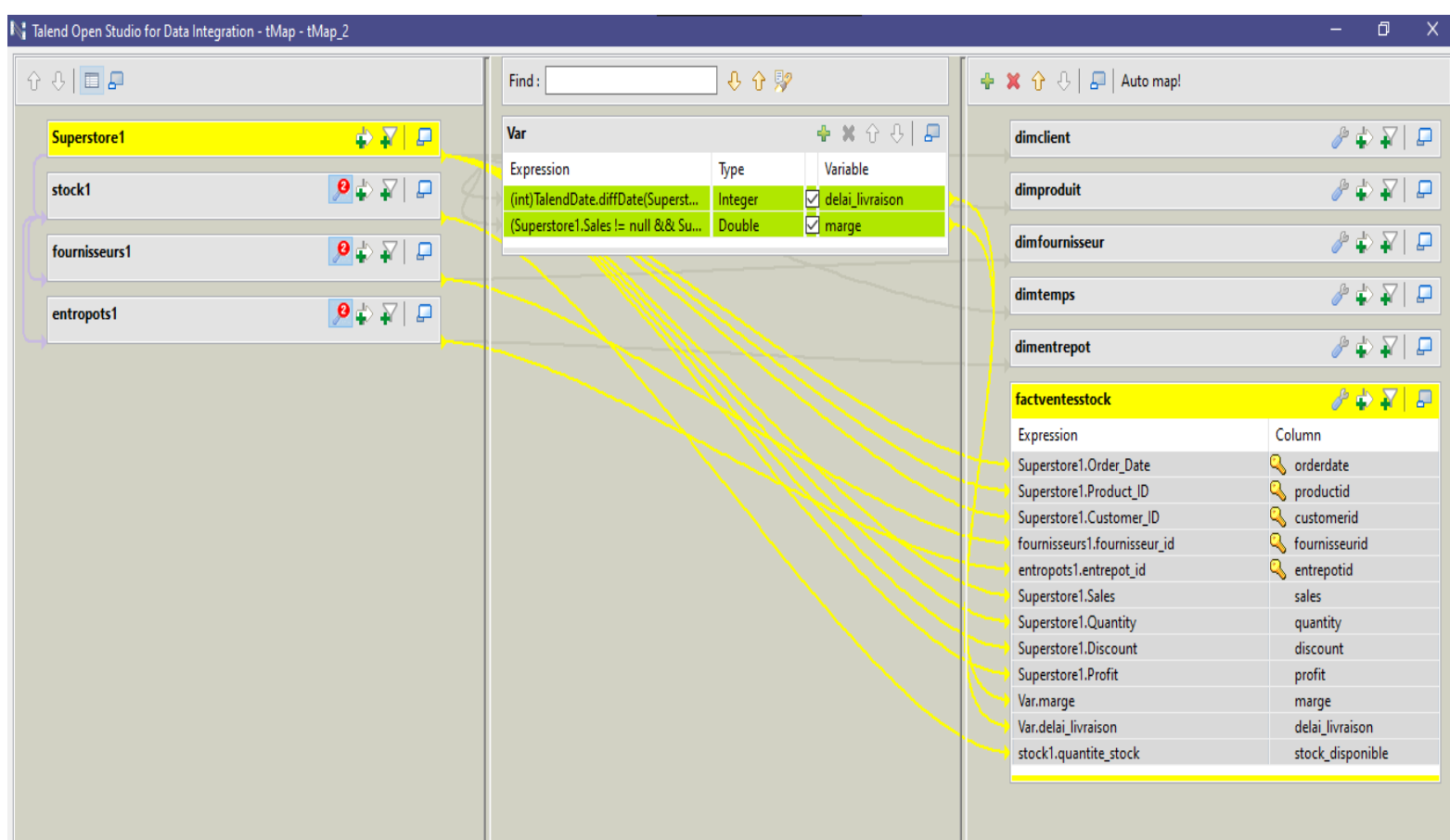
ETL

Voici le schéma global dans Talend, avec l'ajoute de SCD (seulement dans ce cas **dimscdproduit** et **dimscdfournisseur**)



Tmap :

Voici le **tMap** générale :



Dans tMap, voici l'input de fichier CSV **superstore**, tables **stock**, **entrepôts**, **fournisseurs**

Superstore1	
Column	
Row_ID	
Order_ID	
Order_Date	
Ship_Date	
Ship_Mode	
Customer_ID	
Customer_Name	
Segment	
Country	
City	
State	
Postal_Code	
Region	
Product_ID	
Category	
Sub_Category	
Product_Name	
Sales	
Quantity	
Discount	
Profit	

stock1	
Property	Value
Lookup Model	Load once
Match Model	Unique match
Join Model	Inner Join
Store temp data	false
Expr. key	Column
Superstore1.Product_ID	product_id
	nom_produit
	fournisseur_id
	entrepot_id
	quantite_stock
	date_dernier_reap...

entrepots1	
Property	Value
Lookup Model	Load once
Match Model	Unique match
Join Model	Inner Join
Store temp data	false
Expr. key	Column
stock1.entrepot_id	entrepot_id
	nom_entrepot
	region

fournisseurs1	
Property	Value
Lookup Model	Load once
Match Model	Unique match
Join Model	Inner Join
Store temp data	false
Expr. key	Column
stock1.fournisseur_id	fournisseur_id
	nom_fournisseur
	pays
	email_contact

Jointures

Voici les jointures qui j'ai appliquée

Superstore + Stock → via **product_id**

Stock + Fournisseurs → via **fournisseur_id**

Stock + Entrepots → via **entrepot_id**

Et voici les sorties de tMap sous forme les **dimensions** et **tables des faits**

✓ **Dimfournisseur :**

dimfournisseur

Expression	Column
fournisseurs1.fournisseur_id	 fournisseurid
fournisseurs1.nom_fournisseur	nom_fournisseur
fournisseurs1.pays	pays
fournisseurs1.email_contact	email_contact

✓ **Dimproduit :**

dimproduit

Expression	Column
Superstore1.Product_ID	 productid
Superstore1.Product_Name	productname
Superstore1.Sub_Category	subcategoryname
Superstore1.Category	categoryname

✓ **dimclient :**

dimclient

Expression	Column
Superstore1.Customer_ID	 customerid
Superstore1.Customer_Name	customername
Superstore1.Segment	segment
Superstore1.City	city
Superstore1.State	state
Superstore1.Postal_Code	postalcode
Superstore1.Region	region
Superstore1.Country	country

✓ **Dimtemps :**

dimtemps

Expression	Column
Superstore1.Order_Date	 orderdate
TalendDate.getPartOfDate("YEAR", Supers...	annee
Integer.parseInt(TalendDate.formatDate(" ...	mois
((TalendDate.getPartOfDate("MONTH", S...	trimestre
TalendDate.getPartOfDate("DAY_OF_MO...	jour

✓ **Dimentrepot :**

dimentrepot

Expression	Column
entrepots1.entrepot_id	 entrepotid
entrepots1.nom_entrepot	nom_entrepot
entrepots1.region	nom_region

Le tables des faits :

❖ **Factventesstock :**

factventesstock	
Expression	Column
Superstore1.Order_Date	orderdate
Superstore1.Product_ID	productid
Superstore1.Customer_ID	customerid
fournisseurs1.fournisseur_id	fournisseurid
entrepots1.entrepot_id	entrepotid
Superstore1.Sales	sales
Superstore1.Quantity	quantity
Superstore1.Discount	discount
Superstore1.Profit	profit
Var.marge	marge
Var.delai_livraison	delai_livraison
stock1.quantite_stock	stock_disponible

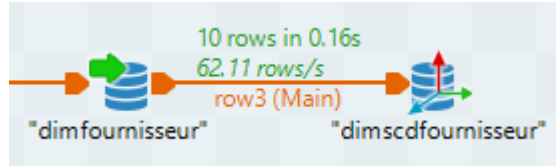
Les variables utilisées dans la table des faits pour calculer le **délai de livraison** et le **stock disponible** et la **marge**

Var		
Expression	Type	Variable
(int)TalendDate.diffDate(Superst...	Integer	☒ delai_livraison
(Superstore1.Sales != null && Su...	Double	☒ marge

Marge = Profit / Sales

Delai_Livraison = Ship_Date – Order_Date

Slowly Changing Dimensions (SCD) :



SCD component editor

filter

Unused

Source keys

fournisseurid

Surrogate keys

name: four_key

creation: Auto increment

complement

Type 0 fields

Type 1 fields

Type 2 fields

email_contact
nom_fournisseur
pays

Versioning

type	name	creation	com
start	scd_start	Job start time	
end	scd_end	NULL	
☐ version	scd_version		
☒ active	scd_active		

Type 3 fields

current value	previous value

OK Cancel

L'affichage avant d'exécuter le job, et avant de changer quelque chose dans les données d'entrer

```
dw_superstore=# select * from dimscdfournisseur;
```

fournisseurid	nom_fournisseur	pays	email_contact	scd_start	scd_end	scd_active	four_key
1	Fournitures USA	Etats-Unis	contact@fournituresusa.com	2025-12-05		t	1
2	Mobilier Europe	France	contact@mobilierfrance.fr	2025-12-05		t	2
3	Papeterie Monde	Allemagne	contact@papeteriemonde.de	2025-12-05		t	3
4	Techno Supplies	Royaume-Uni	contact@technosupplies.co.uk	2025-12-05		t	4
5	Bureautique Inc.	Canada	contact@bureautiqueinc.ca	2025-12-05		t	5
6	Fournitures Asie	Chine	contact@fournituresasie.cn	2025-12-05		t	6
7	Mobilier Afrique	Afrique du Sud	contact@mobilierafrique.co.za	2025-12-05		t	7
8	Papeterie Amérique Latine	Brésil	contact@papeteriebrasil.com	2025-12-05		t	8
9	Techno Supplies	Japon	contact@technosupplies.jp	2025-12-05		t	9
10	Bureautique Europe	Espagne	contact@bureautiqueeurope.es	2025-12-05		t	10

(10 rows)

```
dw_superstore=#
```

Et voici L'affichage après l'exécuter de jobe, et après de changer le pays France de fournisseur d'ID 2 par le pays Maroc dans les données d'entrer

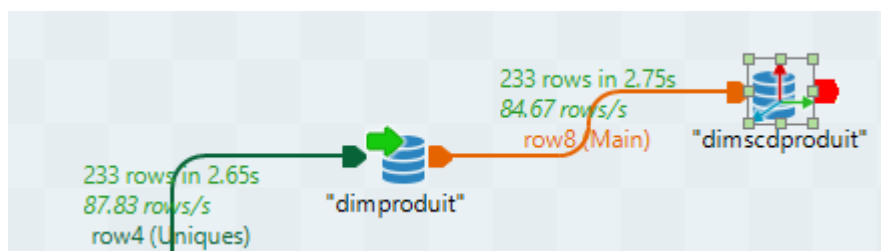
```
dw_superstore=# select * from dimscdfournisseur;
```

fournisseurid	nom_fournisseur	pays	email_contact	scd_start	scd_end	scd_active	four_key
1	Fournitures USA	Etats-Unis	contact@fournituresusa.com	2025-12-05		t	1
3	Papeterie Monde	Allemagne	contact@papeteriemonde.de	2025-12-05		t	3
4	Techno Supplies	Royaume-Uni	contact@technosupplies.co.uk	2025-12-05		t	4
5	Bureautique Inc.	Canada	contact@bureautiqueinc.ca	2025-12-05		t	5
6	Fournitures Asie	Chine	contact@fournituresasie.cn	2025-12-05		t	6
7	Mobilier Afrique	Afrique du Sud	contact@mobilierafrique.co.za	2025-12-05		t	7
8	Papeterie Amérique Latine	Brésil	contact@papeteriebrasil.com	2025-12-05		t	8
9	Techno Supplies	Japon	contact@technosupplies.jp	2025-12-05		t	9
10	Bureautique Europe	Espagne	contact@bureautiqueeurope.es	2025-12-05		t	10
2	Mobilier Europe	France	contact@mobilierfrance.fr	2025-12-05	2025-12-05	f	2
2	Mobilier Europe	Maroc	contact@mobilierfrance.fr	2025-12-05		t	13

(11 rows)

```
dw_superstore=#
```

La même chose pour le scd de dimproduit



SCD component editor

filter

Unused

Source keys

productid

Surrogate keys

name
 creation
 complement

Type 0 fields

Type 1 fields

productname

Type 2 fields

categoryname
subcategoryname

Versioning

type	name	creation	corr
start	scd_start	Job start time	
end	scd_end	NULL	
☐ version	scd_version		
☒ active	scd_active		

Type 3 fields

current value

previous value

OK Cancel

```
dw_superstore=# select * from dimscdproduit;
ERROR: character with byte sequence 0xef 0xbf 0xbd in encoding "UTF8" has no equivalent in encoding "WIN1252"
dw_superstore=# select * from dimscdproduit limit 5;
 prod_key | productid | productname | subcategoryname | categoryname | scd_start | scd_end | scd_active
-----+-----+-----+-----+-----+-----+-----+-----
      1 | FUR-BO-10001798 | Bush Somerset Collection Bookcase | Bookcases | Furniture | 2025-12-05 |  | t
      2 | FUR-CH-10000454 | Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back | Chairs | Furniture | 2025-12-05 |  | t
      3 | OFF-LA-10000240 | Self-Adhesive Address Labels for Typewriters by Universal | Labels | Office Supplies | 2025-12-05 |  | t
      4 | FUR-TA-10000577 | Bretford CR4500 Series Slim Rectangular Table | Tables | Furniture | 2025-12-05 |  | t
      5 | OFF-ST-10000760 | Eldon Fold 'N Roll Cart System | Storage | Office Supplies | 2025-12-05 |  | t
(5 rows)

dw_superstore=#
```

Résultats après la création de Data Warehouse

✓ Dimclient :

```
dw_superstore=# SELECT COUNT(*) as total_client
dw_superstore=# FROM dimclient;
total_client
-----
      633
(1 row)

dw_superstore=# select * from dimclient limit 5;
 customerid | customername | segment | city | state | postalcode | region | country
-----+-----+-----+-----+-----+-----+-----+-----
SO-20335 | Sean O'Donnell | Consumer | Fort Lauderdale | Florida | 33311 | South | United States
AA-10480 | Andrew Allen | Consumer | Concord | North Carolina | 28027 | South | United States
SF-20065 | Sandra Flanagan | Consumer | Philadelphia | Pennsylvania | 19140 | East | United States
SA-20830 | Sue Ann Reed | Consumer | Fresno | California | 93727 | West | United States
EB-14110 | Eugene Barchas | Consumer | San Jose | California | 95123 | West | United States
(5 rows)
```

✓ dimtemps

```
dw_superstore=# select count(*) as t
dw_superstore=# from dimtemps;
t
---
    724
(1 row)

dw_superstore=# select * from dimtemps limit 5;
 orderdate | annee | mois | trimestre | jour
-----+-----+-----+-----+-----
2015-02-21 | 2015 | 2 | 1 | 21
2015-06-20 | 2015 | 6 | 2 | 20
2014-12-02 | 2014 | 12 | 4 | 2
2014-09-21 | 2014 | 9 | 3 | 21
2017-11-03 | 2017 | 11 | 4 | 3
(5 rows)
```

✓ Dimfournisseur :

```
dw_superstore=# select * from dimfournisseur;
```

fournisseurid	nom_fournisseur	pays	email_contact
1	Fournitures USA	Etats-Unis	contact@fournituresusa.com
2	Mobilier Europe	Maroc	contact@mobilierfrance.fr
3	Papeterie Monde	Allemagne	contact@papeteriemonde.de
4	Techno Supplies	Royaume-Uni	contact@technosupplies.co.uk
5	Bureautique Inc.	Canada	contact@bureautiqueinc.ca
6	Fournitures Asie	Chine	contact@fournituresasie.cn
7	Mobilier Afrique	Afrique du Sud	contact@mobilierafrique.co.za
8	Papeterie AmÉrique Latine	BrÚsil	contact@papeteriebrasil.com
9	Techno Supplies	Japon	contact@technosupplies.jp
10	Bureautique Europe	Espagne	contact@bureautiqueeurope.es

(10 rows)

✓ Dimproduit :

```
dw_superstore=# select count(*) as t
dw_superstore=# from dimproduit;
t
-----
233
(1 row)
```

```
dw_superstore=# select * from dimproduit limit 5;
```

productid	productname	subcategoryname	categoryname
OFF-AR-10001868	Prang Dustless Chalk Sticks	Art	Office Supplies
TEC-AC-10004633	Verbatim 25 GB 6x Blu-ray Single Layer Recordable Disc, 3/Pack	Accessories	Technology
OFF-BI-10001078	Acco PRESSTEX Data Binder with Storage Hooks, Dark Blue, 14 7/8" X 11"	Binders	Office Supplies
OFF-PA-10003892	Xerox 1943	Paper	Office Supplies
FUR-FU-10000397	Luxo Economy Swing Arm Lamp	Furnishings	Furniture

(5 rows)

✓ dimentrepot

```
dw_superstore=# select * from dimentrepot;
entrepotid | nom_entrepot | nom_region
-----+-----+-----
          5 | Entrepot Centre | Centre
          2 | Entrepot Est | Est
          3 | Entrepot Sud | Sud
          4 | Entrepot Nord | Nord
          9 | Entrepot Nord | Nord
         10 | Entrepot Centre | Centre
          7 | Entrepot Est | Est
          8 | Entrepot Sud | Sud
          6 | Entrepot Ouest | Ouest
          1 | Entrepot Ouest | Ouest
        100 | Entrepot Global | Global
(11 rows)
```

L'analyse après la création du Data Warehouse

Analyse des ventes

Total des ventes par produit

```
dw_superstore=# SELECT p.productname, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.productname
dw_superstore=# ORDER BY ca_total DESC;
```

productname	ca_total
Lexmark MX611dhe Monochrome Laser Printer	16829.90
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	15610.97
SAFCO Arco Folding Chair	11572.78
Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	10637.53
Hon 4700 Series Mobuis Mid-Back Task Chairs with Adjustable Arms	7582.38
Atlantic Metals Mobile 3-Shelf Bookcases, Custom Colors	7539.71
Bretford CR4500 Series Slim Rectangular Table	7242.78
GBC DocuBind 300 Electric Binding Machine	6785.14
GE 30524EE4	5291.74
Swingline SM12-08 MicroCut Jam Free Shredder	5204.25
Atlantic Metals Mobile 4-Shelf Bookcases, Custom Colors	5184.10
Chromcraft Rectangular Conference Tables	4952.67

Total des ventes par catégorie

```
dw_superstore=# SELECT p.categoryname, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.categoryname
dw_superstore=# ORDER BY ca_total DESC;
categoryname | ca_total
-----+-----
Furniture    | 113967.73
Office Supplies | 79461.90
Technology   | 65929.49
(3 rows)
```

Total des ventes par sous-catégorie

```
dw_superstore=# SELECT p.subcategoryname, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.subcategoryname
dw_superstore=# ORDER BY ca_total DESC;
subcategoryname | ca_total
-----+-----
Chairs          | 48217.88
Bookcases       | 34485.21
Phones          | 28514.63
Machines        | 26302.15
Appliances      | 21555.51
Tables          | 21118.84
Storage         | 20335.36
Binders         | 15385.93
Accessories     | 11112.71
Furnishings     | 10145.80
Art             | 9027.57
Paper           | 8763.19
Envelopes       | 2026.13
Labels          | 1439.29
Fasteners       | 577.81
Supplies        | 351.11
(16 rows)
```

Total des ventes par région

```
dw_superstore=# SELECT c.region, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimclient c ON f.customerid = c.customerid
dw_superstore=# GROUP BY c.region
dw_superstore=# ORDER BY ca_total DESC;
 region | ca_total
-----+-----
 West   | 79459.97
 Central| 71218.75
 East    | 67515.02
 South   | 41165.38
(4 rows)
```

Total des ventes par ville

```
dw_superstore=# SELECT c.city, c.state, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimclient c ON f.customerid = c.customerid
dw_superstore=# GROUP BY c.city, c.state
dw_superstore=# ORDER BY ca_total DESC
dw_superstore=# LIMIT 10;
 city      | state      | ca_total
-----+-----+-----
 New York City | New York   | 21964.75
 Los Angeles  | California | 16939.73
 Houston      | Texas      | 13944.20
 San Antonio  | Texas      | 11042.98
 Seattle      | Washington | 10560.05
 Philadelphia  | Pennsylvania | 10047.82
 San Francisco | California | 8205.02
 Chicago      | Illinois   | 7879.44
 Richmond     | Kentucky   | 5416.37
 Burlington   | Vermont    | 4404.90
(10 rows)
```

Total des ventes par Etat

```
dw_superstore=# SELECT c.state, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimclient c ON f.customerid = c.customerid
dw_superstore=# GROUP BY c.state
dw_superstore=# ORDER BY ca_total DESC;
state | ca_total
-----+-----
California | 41372.70
Texas | 34837.64
New York | 26928.09
Illinois | 14158.85
Washington | 12686.83
Colorado | 11929.59
Ohio | 11898.82
Pennsylvania | 11866.64
Kentucky | 8598.11
Arizona | 8157.19
Wisconsin | 7798.97
North Carolina | 6691.64
New Jersey | 5332.50
```

Total des ventes par segment client

```
dw_superstore=# SELECT c.segment, SUM(f.sales) AS ca_total
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimclient c ON f.customerid = c.customerid
dw_superstore=# GROUP BY c.segment
dw_superstore=# ORDER BY ca_total DESC;
segment | ca_total
-----+-----
Consumer | 141476.20
Corporate | 78692.72
Home Office | 39190.20
(3 rows)
```

Analyse de la rentabilité

La marge totale

```
dw_superstore=# SELECT
dw_superstore=#     SUM(profit) AS profit_total,
dw_superstore=#     SUM(sales) AS ca_total,
dw_superstore=#     ROUND(SUM(profit) * 100.0 / SUM(sales), 2) AS marge_percent_globale
dw_superstore=# FROM factventesstock;
profit_total | ca_total | marge_percent_globale
-----+-----+-----
19271.71 | 259359.12 | 7.43
(1 row)
```

Marge par produit

```
dw_superstore=# SELECT
dw_superstore=#     p.productname,
dw_superstore=#     SUM(f.profit) AS profit_total,
dw_superstore=#     SUM(f.sales) AS ca_total,
dw_superstore=#     ROUND(SUM(f.profit) * 100.0 / NULLIF(SUM(f.sales), 0), 2) AS marge_percent
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.productname
dw_superstore=# HAVING SUM(f.sales) > 0
dw_superstore=# ORDER BY marge_percent DESC;
```

productname	profit_total	ca_total	marge_percent
Prang Dustless Chalk Sticks	18.81	38.97	48.27
Staple envelope	61.41	130.64	47.01
Avery 485	300.33	639.03	47.00
Xerox 232	56.50	120.53	46.88
Array Parchment Paper, Assorted Colors	49.51	106.29	46.58
Xerox 1911	279.74	603.54	46.35
Xerox 205	69.46	150.34	46.20
Xerox 1880	285.65	623.74	45.80
Xerox 1920	40.37	89.70	45.01
Snap-A-Way Black Print Carbonless Ruled Speed Letter, Triplicate	453.76	1016.79	44.63
Poly String Tie Envelopes	30.29	68.13	44.46
Avery 505	165.76	372.96	44.44
Adams Phone Message Book, Professional, 400 Message Capacity, 5 3/6" x 11"	221.85	501.39	44.25
Xerox 1943	328.67	743.43	44.21
Sanford Uni-Blazer View Highlighters, Chisel Tip, Yellow	13.55	30.80	43.99
Globe Weis Peel & Seal First Class Envelopes	164.87	375.73	43.88
Newell Chalk Holder	36.59	83.42	43.86
Easy-staple paper	85.84	196.48	43.69
Avery 511	24.64	56.68	43.47
Xerox 1967	38.87	89.42	43.47
Xerox 1913	375.04	865.49	43.33

Performance des clients et segments

Top clients

```
dw_superstore=# SELECT
dw_superstore=#     c.customername,
dw_superstore=#     c.segment,
dw_superstore=#     c.city,
dw_superstore=#     c.state,
dw_superstore=#     SUM(f.sales) AS ca_total,
dw_superstore=#     SUM(f.quantity) AS quantite_totale,
dw_superstore=#     COUNT(*) AS nb_transactions
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimclient c ON f.customerid = c.customerid
dw_superstore=# GROUP BY c.customername, c.segment, c.city, c.state
dw_superstore=# ORDER BY ca_total DESC
dw_superstore=# LIMIT 10;
```

customername	segment	city	state	ca_total	quantite_totale	nb_transactions
Becky Martin	Consumer	San Antonio	Texas	10755.88	40	8
Quincy Jones	Corporate	Burlington	Vermont	4404.90	5	1
Peter Fuller	Consumer	Richmond	Kentucky	4322.97	15	3
Bart Watters	Corporate	Milwaukee	Wisconsin	3893.94	28	5
Joe Elijah	Consumer	Broomfield	Colorado	3775.78	26	6
Brosina Hoffman	Consumer	Los Angeles	California	3759.36	42	9
Tracy Blumstein	Consumer	Philadelphia	Pennsylvania	3470.27	35	9
Sean Braxton	Corporate	Houston	Texas	3409.09	13	3
Katrina Willman	Consumer	New York City	New York	3085.56	9	2
Sharelle Roach	Home Office	Louisville	Colorado	2659.89	10	2

(10 rows)

Top produit

```
dw_superstore=# SELECT
dw_superstore=#     p.productname,
dw_superstore=#     p.categoryname,
dw_superstore=#     p.subcategoryname,
dw_superstore=#     SUM(f.sales) AS ca_total,
dw_superstore=#     SUM(f.quantity) AS quantite_vendue,
dw_superstore=#     SUM(f.profit) AS profit_total,
dw_superstore=#     ROUND(SUM(f.profit) * 100.0 / NULLIF(SUM(f.sales), 0), 2) AS marge_percent
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.productname, p.categoryname, p.subcategoryname
dw_superstore=# ORDER BY ca_total DESC
dw_superstore=# LIMIT 10;
```

productname	categoryname	subcategoryname	ca_total	quantite_vendue	profit_total	marge_percent
Lexmark MX611dhe Monochrome Laser Printer	Technology	Machines	16829.90	18	-4589.97	-27.27
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	Furniture	Bookcases	15610.97	24	-669.53	-4.29
SAFCO Arco Folding Chair	Furniture	Chairs	11572.78	53	1179.37	10.19
Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	Furniture	Chairs	10637.53	51	1927.45	18.12
Hon 4700 Series Mobuis Mid-Back Task Chairs with Adjustable Arms	Furniture	Chairs	7582.38	24	833.00	10.99
Atlantic Metals Mobile 3-Shelf Bookcases, Custom Colors	Furniture	Bookcases	7539.71	35	780.32	10.35
Bretford CR4500 Series Slim Rectangular Table	Furniture	Tables	7242.78	29	-532.76	-7.36
GBC DocuBind 300 Electric Binding Machine	Office Supplies	Binders	6785.14	23	494.42	7.29
GE 30524EE4	Technology	Phones	5291.74	32	838.84	15.85
Swingline SM12-08 MicroCut Jam Free Shredder	Technology	Machines	5204.25	22	-882.12	-16.95

(10 rows)

Analyse croisée ventes-stock

Produits populaires avec faible stock

```
dw_superstore=# SELECT
dw_superstore=#     p.productname,
dw_superstore=#     SUM(f.sales) AS ca_total,
dw_superstore=#     AVG(f.stock_disponible) AS stock_moyen,
dw_superstore=#     CASE
dw_superstore=#         WHEN AVG(f.stock_disponible) <= 10 THEN 'RUPTURE RISQUE'
dw_superstore=#         WHEN AVG(f.stock_disponible) <= 20 THEN 'STOCK FAIBLE'
dw_superstore=#         ELSE 'STOCK OK'
dw_superstore=#     END AS statut_stock
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.productname
dw_superstore=# HAVING SUM(f.sales) > 10000
dw_superstore=# ORDER BY stock_moyen ASC, ca_total DESC;
```

productname	ca_total	stock_moyen	statut_stock
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish	15610.97	8.0000000000000000	RUPTURE RISQUE
Lexmark MX611dhe Monochrome Laser Printer	16829.90	12.0000000000000000	STOCK FAIBLE
Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	10637.53	26.0000000000000000	STOCK OK
SAFCO Arco Folding Chair	11572.78	34.0000000000000000	STOCK OK

(4 rows)

Produits populaires avec faible marge

```
dw_superstore=# SELECT
dw_superstore=#     p.productname,
dw_superstore=#     SUM(f.sales) AS ca_total,
dw_superstore=#     SUM(f.profit) AS profit_total,
dw_superstore=#     ROUND(SUM(f.profit) * 100.0 / NULLIF(SUM(f.sales), 0), 2) AS marge_percent
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimproduit p ON f.productid = p.productid
dw_superstore=# GROUP BY p.productname
dw_superstore=# HAVING SUM(f.sales) > 5000 AND
dw_superstore=#     ROUND(SUM(f.profit) * 100.0 / NULLIF(SUM(f.sales), 0), 2) < 10
dw_superstore=# ORDER BY marge_percent ASC, ca_total DESC;
productname | ca_total | profit_total | marge_percent
-----+-----+-----+-----
Lexmark MX611dhe Monochrome Laser Printer | 16829.90 | -4589.97 | -27.27
Swingline SM12-08 MicroCut Jam Free Shredder | 5204.25 | -882.12 | -16.95
Bretford CR4500 Series Slim Rectangular Table | 7242.78 | -532.76 | -7.36
Riverside Palais Royal Lawyers Bookcase, Royale Cherry Finish | 15610.97 | -669.53 | -4.29
Atlantic Metals Mobile 4-Shelf Bookcases, Custom Colors | 5184.10 | -126.43 | -2.44
GBC DocuBind 300 Electric Binding Machine | 6785.14 | 494.42 | 7.29
(6 rows)
```

Impact des remises sur le profit et la marge

Analyse par niveau de remise

```
dw_superstore=# SELECT
dw_superstore=#     CASE
dw_superstore=#         WHEN discount = 0 THEN '0%'
dw_superstore=#         WHEN discount <= 0.1 THEN '1-10%'
dw_superstore=#         WHEN discount <= 0.2 THEN '11-20%'
dw_superstore=#         WHEN discount <= 0.3 THEN '21-30%'
dw_superstore=#         WHEN discount <= 0.4 THEN '31-40%'
dw_superstore=#         ELSE '>40%'
dw_superstore=#     END AS tranche_remise,
dw_superstore=#     COUNT(*) AS nb_transactions,
dw_superstore=#     SUM(sales) AS ca_total,
dw_superstore=#     SUM(profit) AS profit_total,
dw_superstore=#     ROUND(AVG(discount) * 100, 2) AS remise_moyenne,
dw_superstore=#     ROUND(SUM(profit) * 100.0 / NULLIF(SUM(sales), 0), 2) AS marge_moyenne
dw_superstore=# FROM factventesstock
dw_superstore=# GROUP BY
dw_superstore=#     CASE
dw_superstore=#         WHEN discount = 0 THEN '0%'
dw_superstore=#         WHEN discount <= 0.1 THEN '1-10%'
dw_superstore=#         WHEN discount <= 0.2 THEN '11-20%'
dw_superstore=#         WHEN discount <= 0.3 THEN '21-30%'
dw_superstore=#         WHEN discount <= 0.4 THEN '31-40%'
dw_superstore=#         ELSE '>40%'
dw_superstore=#     END
dw_superstore=# ORDER BY remise_moyenne;
tranche_remise | nb_transactions | ca_total | profit_total | remise_moyenne | marge_moyenne
-----+-----+-----+-----+-----+-----
0% | 713 | 107033.95 | 28779.35 | 0.00 | 26.89
1-10% | 14 | 10390.02 | 1976.19 | 10.00 | 19.02
11-20% | 553 | 88682.13 | 9727.11 | 19.95 | 10.97
21-30% | 27 | 14689.50 | -1290.80 | 30.00 | -8.79
31-40% | 24 | 20110.79 | -3937.75 | 39.33 | -19.58
>40% | 122 | 18452.73 | -15982.39 | 71.02 | -86.61
(6 rows)
```

Corrélation remise-profit par produit

dw_superstore=# SELECT
dw_superstore-# p.productname,
dw_superstore-# ROUND(AVG(f.discount) * 100, 2) AS remise_moyenne,
dw_superstore-# SUM(f.quantity) AS quantite_vendue,
dw_superstore-# SUM(f.sales) AS ca_total,
dw_superstore-# SUM(f.profit) AS profit_total,
dw_superstore-# ROUND(SUM(f.profit) * 100.0 / NULLIF(SUM(f.sales), 0), 2) AS marge_percent,
dw_superstore-# ROUND(CORR(f.discount, f.profit)::numeric, 4) AS correlation_remise_profit
dw_superstore-# FROM factventesstock f
dw_superstore-# JOIN dimproduit p ON f.productid = p.productid
dw_superstore-# GROUP BY p.productname
dw_superstore-# HAVING COUNT(*) > 10
dw_superstore-# ORDER BY correlation_remise_profit DESC;

productname	remise_moyenne	quantite_vendue	ca_total	profit_total	marge_percent	correlation_remise_profit
Xerox 1916	8.57	52	1004.64	415.27	41.34	0.2798
Newell 312	16.36	49	234.77	34.46	14.68	-0.3396
Newell 333	5.45	33	86.74	18.84	21.72	-0.5153
Poly String Tie Envelopes	7.27	35	68.13	30.29	44.46	-0.6231
Howard Miller 13-3/4" Diameter Brushed Chrome Round Wall Clock	11.67	44	1438.13	235.17	16.35	-0.6722
Pressboard Covers with Storage Hooks, 9 1/2" x 11", Light Blue	45.00	44	102.62	-11.87	-11.57	-0.7230
Hon Deluxe Fabric Upholstered Stacking Chairs, Rounded Back	14.17	51	10637.53	1927.45	18.12	-0.7770
Avery Binding System Hidden Tab Executive Style Index Sets	34.55	36	214.57	47.86	22.31	-0.8312
Avery Self-Adhesive Photo Pockets for Polaroid Photos	53.64	41	143.01	-2.19	-1.53	-0.8592
SAFCO Arco Folding Chair	20.00	53	11572.78	1179.37	10.19	-0.8849
Wilson Jones Leather-Like Binders with Dubllock Round Rings	35.00	46	238.33	25.49	10.70	-0.8886
SanDisk Ultra 32 GB MicroSDHC Class 10 Memory Card	10.91	44	884.00	38.02	4.30	-0.9182

(12 rows)

Analyse des livraisons et commandes

Délai moyen de livraison

dw_superstore=# SELECT
dw_superstore-# ROUND(AVG(delai_livraison), 1) AS delai_moyen_jours,
dw_superstore-# MIN(delai_livraison) AS delai_min,
dw_superstore-# MAX(delai_livraison) AS delai_max,
dw_superstore-# PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY delai_livraison) AS delai_median
dw_superstore-# FROM factventesstock
dw_superstore-# WHERE delai_livraison IS NOT NULL;

delai_moyen_jours	delai_min	delai_max	delai_median
3.9	0	7	4

(1 row)

Délai par région

```
dw_superstore=# SELECT
dw_superstore=#     c.region,
dw_superstore=#     ROUND(AVG(f.delai_livraison), 1) AS delai_moyen,
dw_superstore=#     COUNT(*) AS nb_commandes
dw_superstore=# FROM factventesstock f
dw_superstore=# JOIN dimclient c ON f.customerid = c.customerid
dw_superstore=# WHERE f.delai_livraison IS NOT NULL
dw_superstore=# GROUP BY c.region
dw_superstore=# ORDER BY delai_moyen DESC;
 region | delai_moyen | nb_commandes
-----+-----+-----
South   |          4.1 |          234
West    |          4.1 |          434
Central |          3.9 |          366
East    |          3.7 |          419
(4 rows)
```